

DETAILED DESCRIPTION OF THE INVENTION — MULTI-NODE SYSTEM INTEGRATION AND CROSS-DEVICE CORRELATION

In various embodiments, the intraluminal physiologic monitoring device may operate independently or as a functional node within a distributed physiologic sensing ecosystem comprising multiple body-positioned monitoring devices.

The disclosed system architecture may include external wearable or body-associated monitoring nodes configured to measure complementary physiologic conditions. Such nodes may include external ring-form devices, body-worn patches, wrist-mounted devices, chest-positioned monitors, handheld scanning devices, or other physiologic sensing platforms.

Data generated by the intraluminal monitoring device may be temporally synchronized with data obtained from other nodes to enable contextual interpretation of physiologic states. Cross-device correlation may allow identification of relationships between localized pelvic physiology and broader systemic physiologic conditions including cardiovascular context, motion state, thermal regulation, or behavioral activity.

System-level processing may perform multi-node fusion using statistical modeling, signal alignment, machine learning inference, or rule-based computational frameworks. Fusion methods may generate composite physiologic profiles derived from coordinated sensing across multiple regions of the body.

In certain embodiments, cross-device analysis may enable compatibility modeling, comfort alignment evaluation, adaptive calibration between devices, or prediction of physiologic response based on multi-source inputs. The system may generate comparative metrics, relational indices, or synchrony measures representing coordinated physiologic behavior across users or across device nodes.

Multi-node integration may further support adaptive system behavior in which operational parameters of one device are modified based on information received from another device. Examples may include dynamic sampling adjustment, calibration refinement, contextual alerts, or guided measurement workflows.

Communication between nodes may occur through wireless protocols including Bluetooth®, ultra-wideband, near-field communication, Wi-Fi, or other wired or wireless communication architectures. Data exchange may be mediated through companion computing devices or cloud-based coordination systems.

The distributed ecosystem described herein enables scalable expansion through addition of future sensing devices without modification of core system architecture. Alternative node configurations, communication methods, synchronization techniques, and computational models capable of achieving substantially similar functional outcomes are considered within the scope of the disclosure.